SMART INDIA HACKATHON 2026

Problem Statement ID - 26231

Problem Statement Title -Digital Companion for Field Drug Testing

Theme - MedTech / BioTech / HealthTech

PS Category - Software

Team ID - K26234

Team Name - HexTech

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PROPOSED SOLUTION

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Problem

- Officers mainly judge the kit's colour change by looking at it, so lighting and personal judgement can affect the result.
- There is no standard colour value to compare the test result with.
- The test does not always leave a clear digital record of who performed it, where, and when, making it harder to verify later.

Solution

- Check the Test - The mobile app captures the colour change and helps check the result.
- Record the Test - Saves the result along with the time, location, and Inspector ID.
- Manage the Records - The admin dashboard manages Inspectors, test records, evidence, and reports.

User Experience

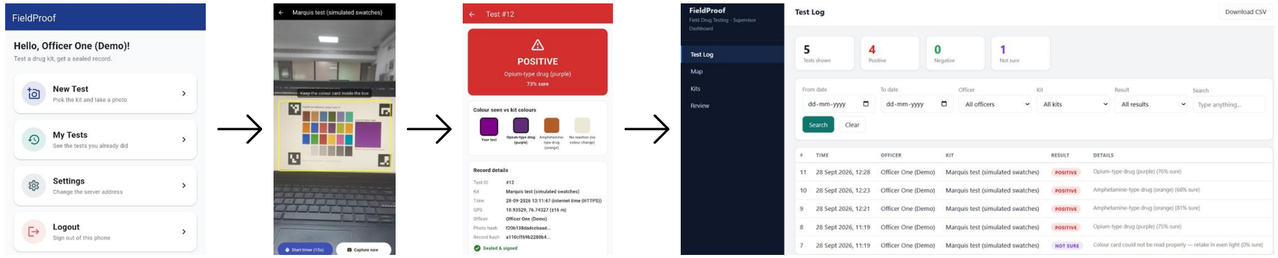
- **4 steps** - pick kit → photo inside the box → result → sealed. Result ~2.5 s after the photo.
- **Bad photo?** - App says NOT SURE with the reason (blur, glare, card not read) and asks to retake.
- **Easy to read** - big result word + colour (not colour alone), simple English. Next: timed test with 10 officers.

Prototype Demo Video: Prototype Video Link

Our Approach

Test Kit → Take Photo → Check Colour → Get Result → Save Proof → Admin Review

Prototype Flow - real screenshots of our working prototype



Test Log (right): search + filter by date, officer, kit and result; export as CSV, PDF + QR.

Novelty

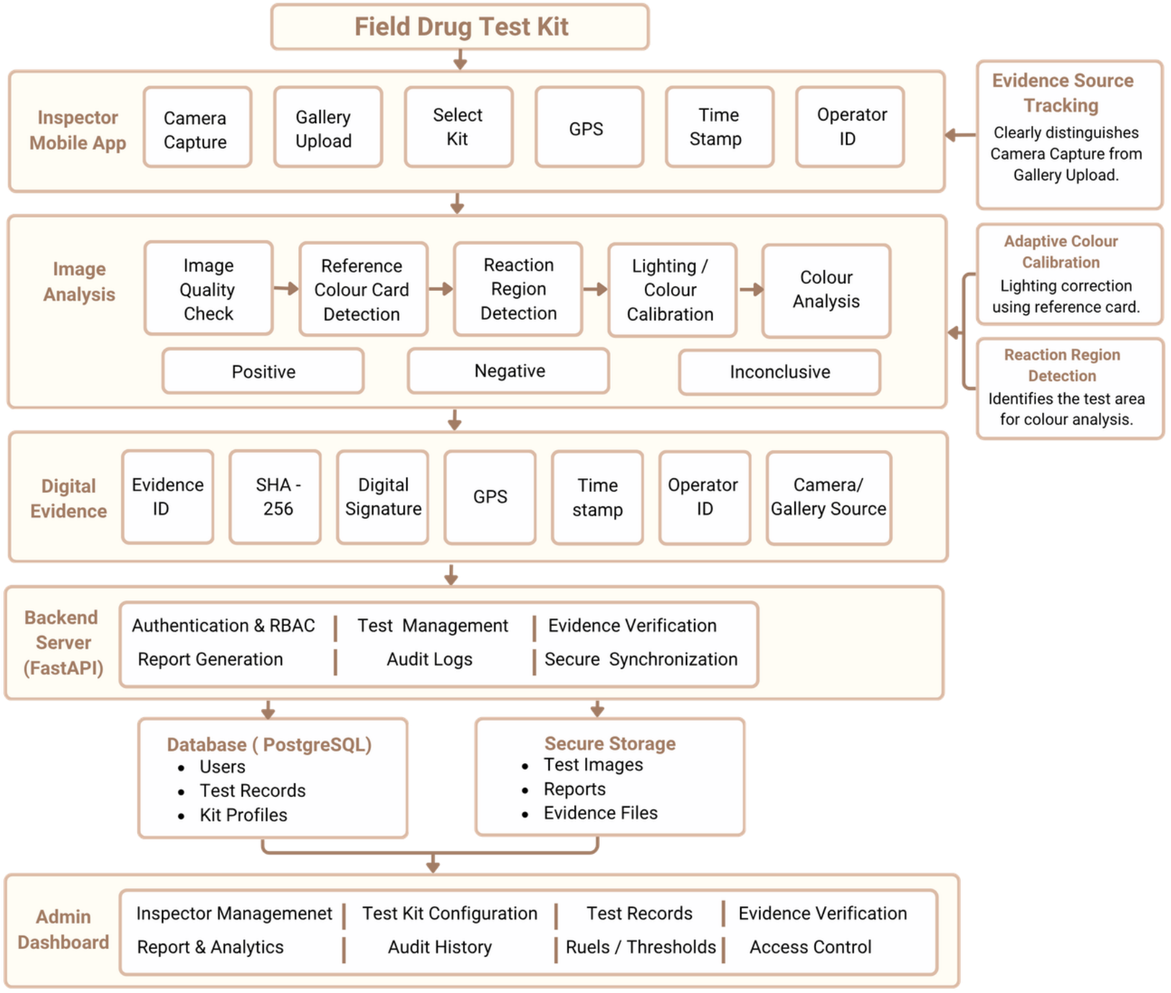
- **Handles Lighting Variations** - Uses a low-cost colour card to improve colour detection under different lighting.
- **Handles Unclear Results** - Blurry, shiny, or unclear images are marked INCONCLUSIVE instead of giving a wrong result.
- **Creates Verifiable Evidence** - Securely records the photo, GPS, time, and Inspector ID, with QR-based verification.
- **Supports Multiple Kits** - New kit colour profiles can be added from the Admin website without changing the app.

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TECHNICAL APPROACH

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System Architecture

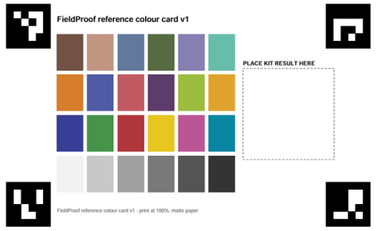


Our Goals - Measured Results

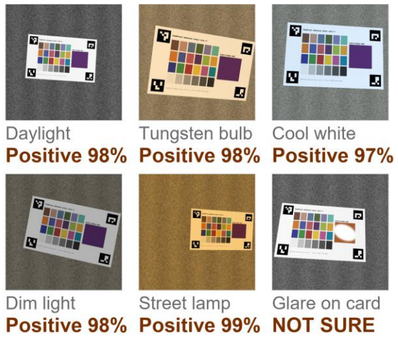
- Testing: 37 kit photos across 5 lighting conditions, 6 real phone photos.
- Accuracy: 25/25 controlled tests correct; 5/6 real photos correct, with 1 marked “NOT SURE”.
- Colour Check: Used ΔE ≤ 14, 4-point colour matching, card-fit and glare thresholds.
- Speed: Result secured in 2.5 s typically, 5 s maximum.
- Tamper Test: All 4/4 modified records were detected successfully.

Kit colours are simulated swatches (test photos + card on a screen), real kits need an FSL calibration trial.
Next step (not done yet): real-kit trial with an FSL, in lab + field light.

Reference Card



Proof of Results



6/6 correct · test run 28 Sep 2026 · simulated swatches

Technology Stack

- Flutter (Dart)
- FastAPI (Python)
- OpenCV + NumPy
- CIELAB + ΔE2000
- PostgreSQL + SQLAlchemy
- React + Vite
- JWT + bcrypt
- SHA-256
- Ed25519
- NTP
- Leaflet + OpenStreetMap
- ReportLab

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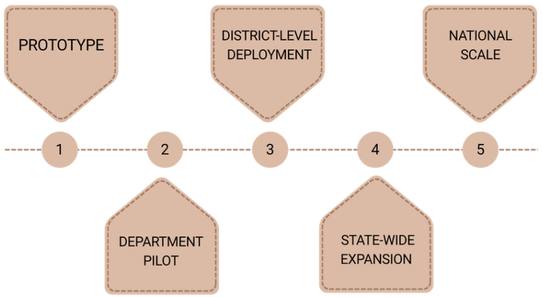
FEASIBILITY AND VIABILITY

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Feasibility & Sustainability

- **Technical** - Core app tested with OpenCV-based colour analysis.
- **User-Friendly** - Capture one photo and get a result within seconds.
- **Low Cost** - Uses an existing Android phone, a reusable ~₹10 colour card, and open-source tools.
- **Operational** - Simple photo-based workflow with minimal training required.
- **Scalable** - Add new test-kit colour profiles through the Admin website without updating the app.
- **Sustainable** - Government-cloud hosting, regular profile updates, and secure data storage support long-term use.

Phased Implementation & Scalability



Pilot Plan (proposed)

- **Where** - 1 district in Tamil Nadu, with local police + nearest FSL
- **Who** - 20 field officers for 3 months
- **Check** - every field result re-tested in the FSL lab
- **Success** - 95%+ match with lab, NOT SURE under 10%, every record verifies
- **Then** - district → state → national (phases on the left)
- **Cost** - about ₹10,000 for 3 months: cards ₹200 + server ~₹3,000/month (estimate)

Potential Challenges & Strategies

Real-world problem	What our app does
Bad light (street lamp / torch)	Colour card fixes the colours
Shiny or blurry photo, card missing	Says NOT SURE and asks to retake
Colour is in between two results	Says NOT SURE; senior officer checks
Fake photo or edited record	In-app camera only + photo fingerprint
Different kits and phones	Kit settings + colours fixed every photo

Cost & Deployment Plan

COST	HOSTING	MAINTENANCE	DEPLOYMENT	SCALE
Uses an existing phone, ₹10 reusable colour card, and open-source software.	Runs on government cloud for the app, database, and admin dashboard.	Regular security updates and easy kit-profile updates as needed.	Start with a pilot, test in the field, then expand deployment.	New test kits can be added through the website without updating the app.

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IMPACT AND BENEFITS

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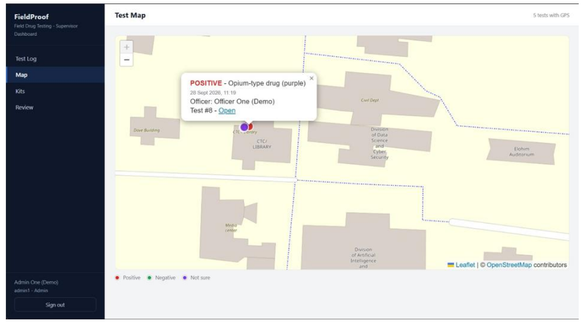
Inspector Benefits

- Faster, more consistent testing
- Less manual colour checking
- INCONCLUSIVE when unclear
- Easy digital test records
- Quick access to past test records

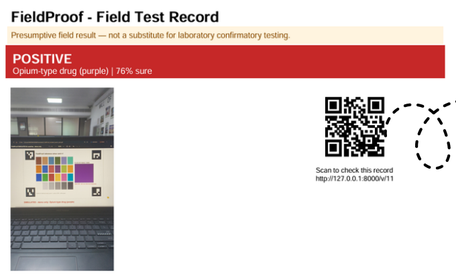
Organisation Benefits

- Proof of every test: time, place, officer & photo
- Secure and verifiable records
- Less paperwork with PDF & QR
- Easy search, reports and monitoring
- Better field-test documentation

Map - Geo-Tagged test result



QR-Based Record Verification



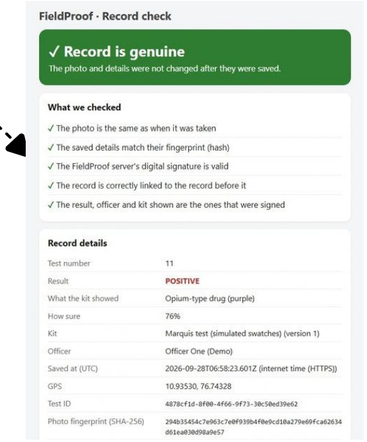
FieldProof - Field Test Record

Presumptive field result — not a substitute for laboratory confirmatory testing.

POSITIVE
Opium-type drug (purple) | 76% sure

Scan to check this record
http://127.0.0.1:8000/v/11

Test number	11
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Saved at (UTC)	2026-09-28T06:58:23.601Z (time from: internet time (HTTPS))
Phone time	2026-09-28T12:28:21+05:30 (difference: -2.6 s)
GPS location	10.935299, 76.743282 (±14.791999816894531 m)
Officer	Officer One (Demo) (user id 1)
Device ID	android-1e42a9c17070165 6ac8a22b0ca44a6e0
Kit	Marquis test (simulated swatches) [DEMO-MARQUIS-SIM], profile version 1
Lab sample ID	-
Review status	NOT_NEEDED



FieldProof · Record check

✓ Record is genuine
The photo and details were not changed after they were saved.

What we checked
- ✓ The photo is the same as when it was taken
- ✓ The saved details match their fingerprint (hash)
- ✓ The FieldProof server's digital signature is valid
- ✓ The record is correctly linked to the record before it
- ✓ The result, officer and kit shown are the ones that were signed

Record details

Test number	11
Result	POSITIVE
What the kit showed	Opium-type drug (purple)
How sure	76%
Kit	Marquis test (simulated swatches) (version 1)
Officer	Officer One (Demo)
Saved at (UTC)	2026-09-28T06:58:23.601Z (internet time (HTTPS))
GPS	10.93530, 76.74328
Test ID	4878cf1d-8f00-4f66-9f73-30c50ed39e62
Photo fingerprint (SHA-256)	294b35454c7e963c7e0f939b4f0e9cd10a279e69fca62634d61ea030d98a9e57

Privacy & Retention (DPDP 2023)

- **Data** - no biometrics, no suspect name (only a sample code). Officer ID, GPS, time kept as evidence.
- **Who sees** - officer: own tests; supervisor/admin: all; public QR: check page (hiding officer name planned).
- **Law** - police work needs no consent (s.7(c), s.17(1)(c)) but data must be kept safe (s.8(5)).
- **Keep** - till case + appeal end, then photo removed (planned).

Tamper Test - real run on our saved records

Test #10 `prev 2ece5c80` `hash 6b1b4e7f` → **Test #11** `prev 6b1b4e7f` `hash 03dd9f0c` → **Test #12** `prev 03dd9f0c` `hash a110cff6`

Ed25519 signature on #11 (key ed25519-1)
`JRIdu6ERgadC+2n6chKfUStpnyNOzHkswJH6iSr4...`

Photo SHA-256
`294b35454c7e963c7e0f939b4f0e9cd10a279e69...`

Each test stores the hash of the test before it, so edits break the chain.

What we changed (#11)	Check page	Failed check
Nothing (original)	GENUINE	none (5/5 pass)
1 byte of the photo	TAMPERED	photo
Result inside sealed copy	TAMPERED	hash, signature, fields
Result in database row	TAMPERED	fields
The signature	TAMPERED	signature

Real run on our saved records, 29 Sep 2026. Changes made in memory only.

Chain of Custody & Court

- **Key** - Ed25519 key only on the server, never in the app; anyone checks with the public key. Next: hardware key box (HSM) + officer DSC.
- **Check** - QR page re-computes the photo hash + record hash, verifies the signature and the chain (see Tamper Test).
- **Court** - gives the SHA-256 hash for the BSA 2023 s.63 certificate; e-Sakshya / CCTNS linking is planned.
- **Offline (design)** - test saved on phone with its hash, sent later; same test sent twice is saved once; phone lost: admin blocks the login (planned).

Future Scope

- Offline Mode - Test even without internet and sync data later.
- Phone Security - Use the phone's security chip to sign each test.
- Police System Integration - Connect with e-Sakshya and CCTNS.
- Lab Comparison - Compare field results with lab results to improve accuracy.
- Regional Languages - Support Hindi and regional languages with voice assistance.

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RESEARCH AND REFERENCES

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What they found	How we use it	Research
IEEE EMBC 2016 Phone camera colours change with light and phone model; reference colours can help handle this.	Our ₹10 colour card fixes colours in every test	IEEE EMBC 2016 – Illumination and device independence for colorimetric detection of urinary biomarkers with smartphone
IEEE TPS-ISA 2024 Digital evidence must maintain integrity and show how it was handled.	Photo fingerprint + signature + chained log	IEEE TPS-ISA 2024 – Digital Evidence Chain of Custody: Navigating New Realities of Digital Forensics
Forensic Science International 2014 A mobile phone camera can measure drug-test colours using RGB values.	Our app reads the test colour as a number	Forensic Science International 2014 – Real-time quantitative colorimetric test for methamphetamine detection using digital and mobile phone technology
Analytical Chemistry 2016 By-eye colour interpretation is subjective; smartphone image analysis can provide objective analysis.	App gives POSITIVE / NEGATIVE / NOT SURE	Analytical Chemistry 2016 – Objective Method for Presumptive Field-Testing of Illicit Drug Possession Using Centrifugal Microdevices and Smartphone Analysis
Digital Discovery 2023 Computer vision can quantify and compare colour changes that may look similar by eye.	App shows photo colour next to kit colours	Digital Discovery 2023 – Teaching old presumptive tests new digital tricks with computer vision